



Sex Disparities in Mortality and Causes of Death in the Netherlands: An Exploration of Historical Amsterdam Data, 1856–1875

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Research Problem

What we know

- Modern demography emphasises the female survival advantage over male counterparts: Females have had higher life expectancy than males for the last 100 years.

But Does Historical Data show a female-male survival advantage?

- Is the female-male survival advantage uniformly present across all age groups and years in 19th-century Amsterdam, Netherlands(1856-1875)?
- What explains the differences in female-male survival across ages and time in the 19th Century?

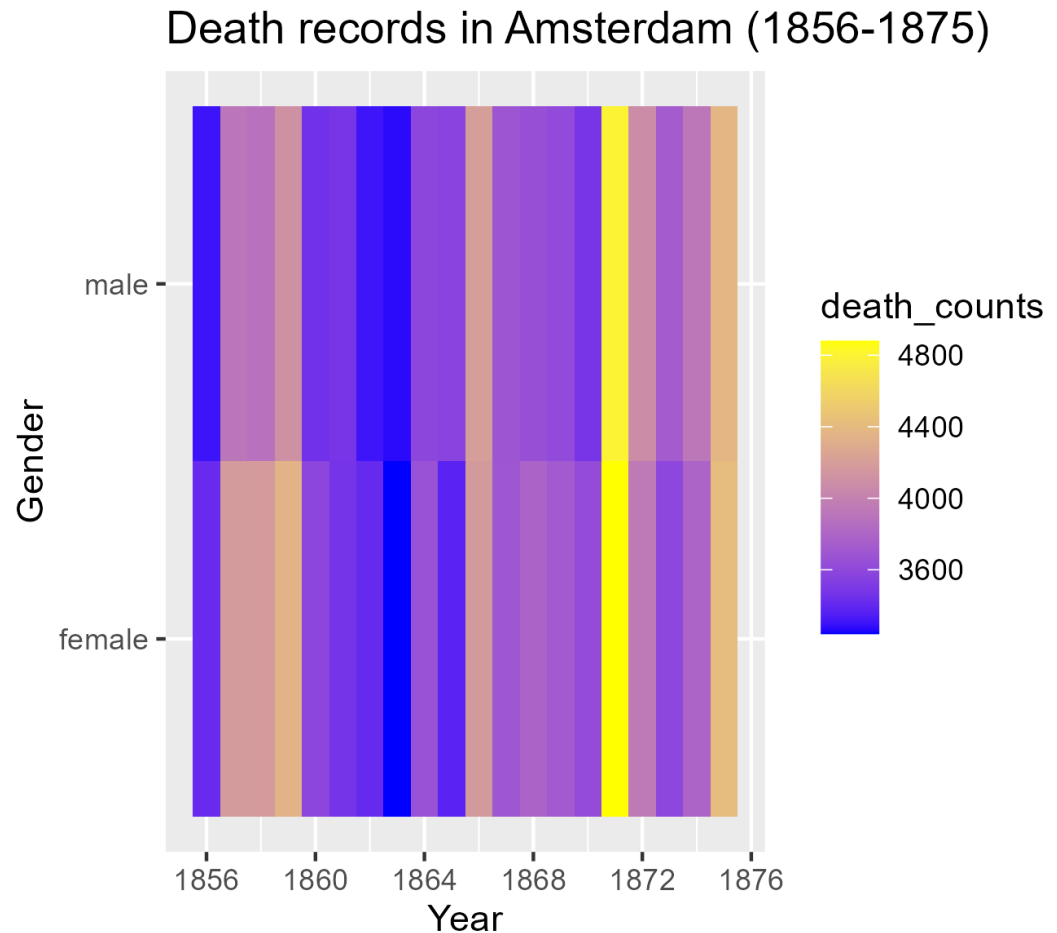
- **Data**
 - Amsterdam Cause of Death (ACD)
 - Amsterdam Population data [comparing age groups in the censuses and in the paper of Marco van Leeuwen]
- **Data description**
 - year
 - month
 - age-year,
 - gender,
 - ICD10h1
- **Type of data**
 - Amsterdam historical data (1856-1875).
- **Source of data**
 - GREAT LEAP

Analytical Techniques

- Descriptive statistics
 - Distribution of death by gender
- Multinomial logistic regression
 - Relative Risk Ratio for women vs men.
- Decomposition analysis of mortality by Gender (Kitagawa, 1955 method).
 - Direct effect: variations in age-specific risks
 - Structure effect: shifts in population structure.

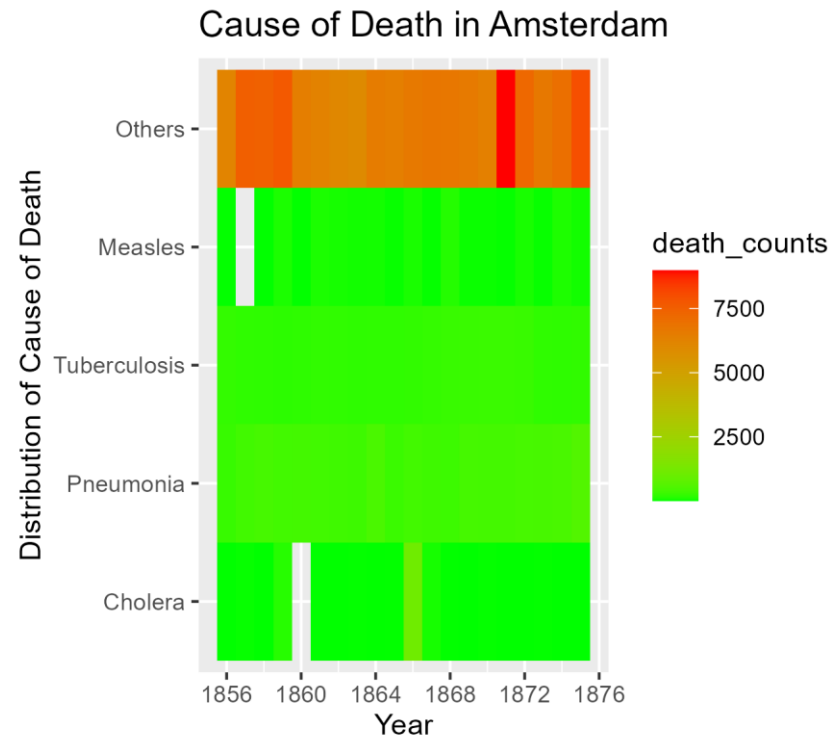
Main Results

- The distribution of deaths is **almost identical for both men and women**.
- **A spike in deaths** for both men and women (1871-1872) may be due to an epidemic.



Causes of death in Amsterdam (1856–1875)

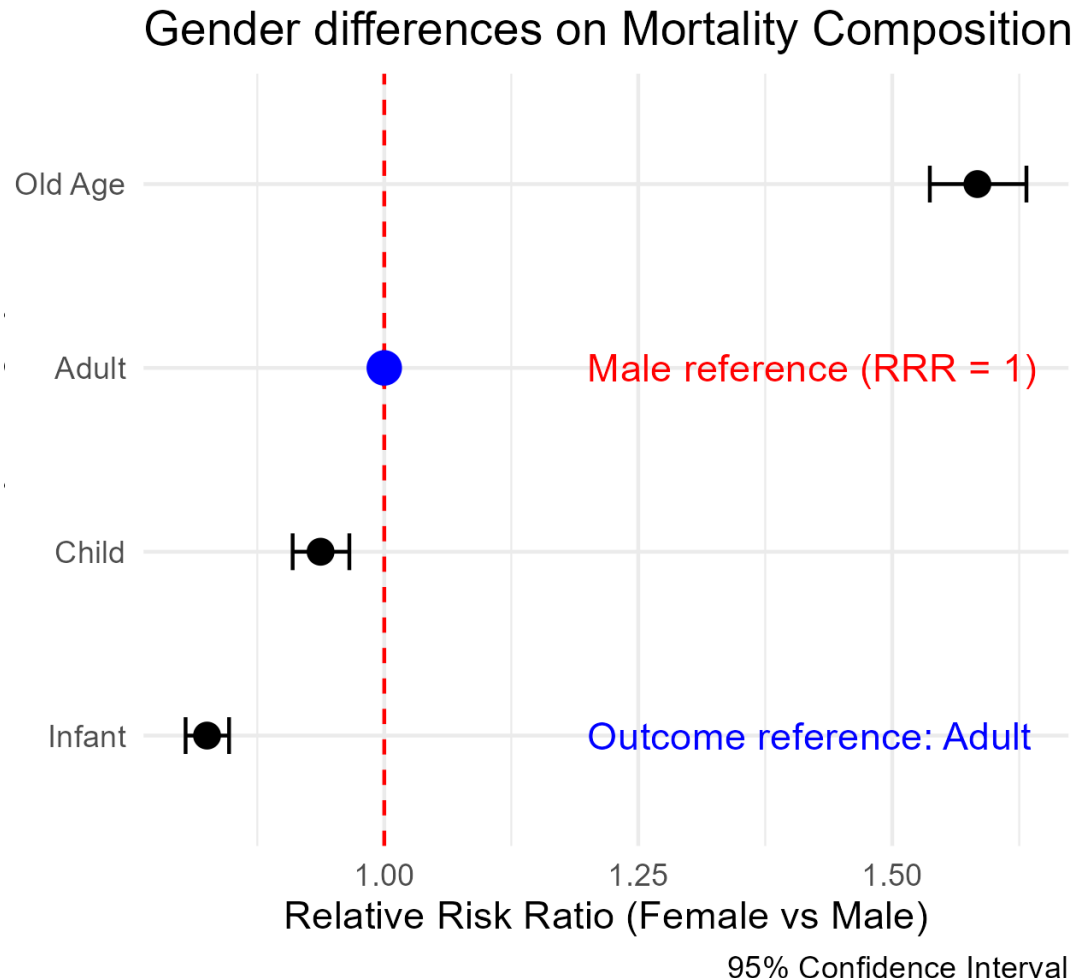
- **Other causes** of death were dominant.
- Epidemic spikes of Cholera and Measles (**short spikes**).



Multinomial Logistic Regression

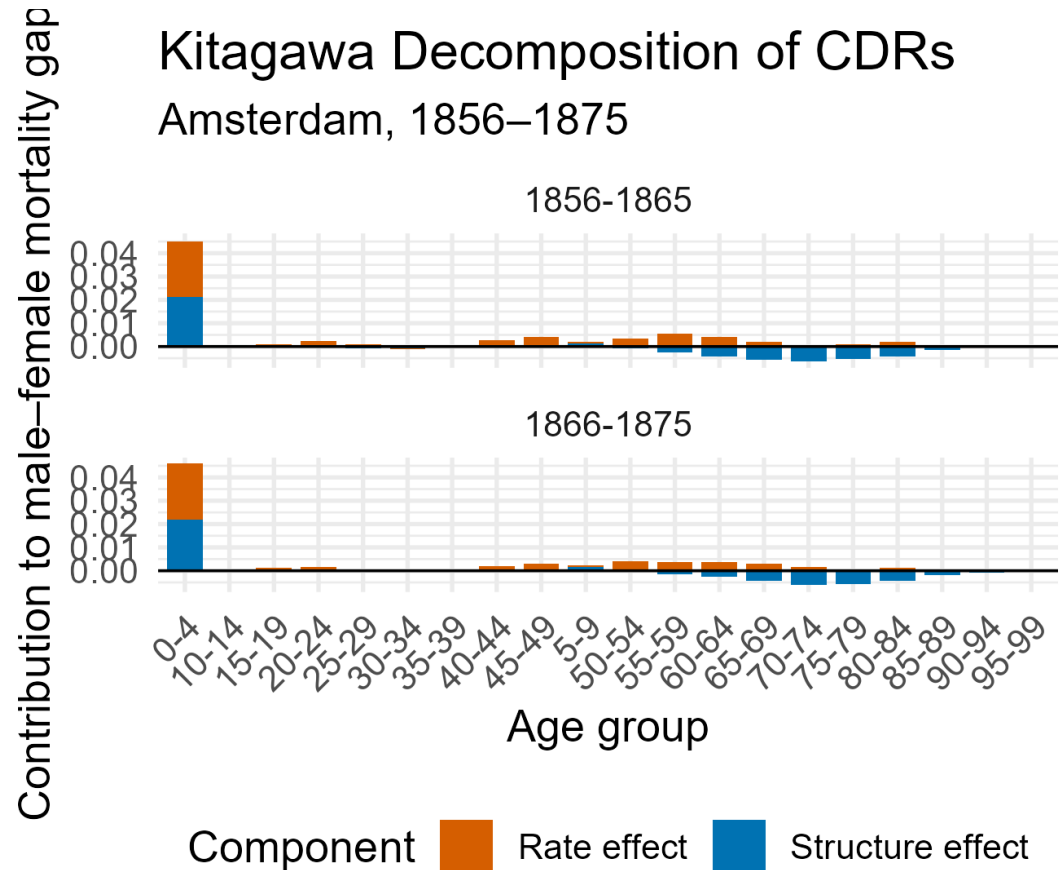
Women exhibit a **high survival advantage** as infants and children compared to adults.

However, Women have a **higher relative risk ratio at old age** than at the adult stage.



Decomposition of Gender Differences in Mortality

Mortality differences are largely driven by **age-specific mortality differences** rather than demographic composition.



CDRs=Crude Death Rates

Conclusions

- The existence of a female advantage **in the mid-19th century** suggests that this pattern is **not purely a modern** phenomenon.
- However, the dominance of infant and child mortality in the 19th century **differs from the contemporary situation**, where adult and older-age mortality plays a larger role in gender differences in mortality.
- As a limitation, the study used the **population estimates** of 1869 and 1879 as a proxy population at risk. Also, the dataset comprised of only death records and **no survivor records**.

END

Thank you for your attention